

Simone Saitta, PhD

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RESEARCH EXPERIENCE

Assistant Professor in Cardiovascular Image Analysis

May 2024 - ongoing

Amsterdam UMC & University of Amsterdam

Amsterdam, The Netherlands

Assistant Professor (Junior)

Nov. 2023 – Apr. 2024

Politecnico di Milano

Milan, Italy

Postdoctoral Research Fellow

May 2023 – Nov. 2023

Politecnico di Milano

Milan, Italy

- Implemented and trained of graph neural networks for fast surrogate modeling of cardiovascular biomechanics.

PhD Student

Nov. 2019 – May 2023

Thesis: Data-driven approaches for quantification of complex biomarkers from medical imaging

Politecnico di Milano

Milan, Italy

- Developed, trained, and applied deep neural networks for automatic 3D segmentation of medical images.
- Integrated deep neural networks within computational pipelines for mixed-reality applications in neurosurgery.
- Developed deep neural network-based methods for unsupervised 4D reconstruction of fluid flow from MRI.
- Developed of a 4D variational data assimilation framework for blood flow dynamics in cerebral aneurysms
- Gained programming experience with the FEniCS API for PDE-constrained optimization using an adjoint solver.

Visiting PhD Student

Mar. 2022 – Aug. 2022

University of Cambridge – Department of Applied Mathematics and Theoretical Physics

Cambridge, UK

- Increased knowledge of inverse problems in medical imaging applications
- Developed an unsupervised learning approach for denoising and super-resolution of flow-encoded MRI.

Consulting

2021 – 2023

Politecnico di Milano & SAATI S.p.A.

Milan, Italy

- Developed user defined subroutines for particle laden numerical simulations of fluid flow.
- Numerical modeling of multi-phase and multi-scale flows in complex geometries using a volume-of-fluid method

Research Engineer

Jan. 2019 – Aug. 2019

National University of Singapore

Singapore

- Developed an automatic image registration computational framework based on iterative optimization methods.
- Implemented automatic multi-view 4D image registration using python and simple-elastix.
- Gained experience with python and C++ cross-platform libraries for object-oriented programming.

Research Assistant

May 2018 – Dec. 2018

Imperial College London

London, UK / Shanghai, China

- Optimized of a finite element-based framework for non-invasive pressure calculation from MRI
- Implemented of a user-friendly GUI for medical image processing in MATLAB
- CFD simulations of blood flow dynamics in aortic dissections using open-source finite element software

QUALIFICATIONS

PhD in Bioengineering (*cum Laude*)

May 2023

Politecnico di Milano

Milan, Italy

Thesis: "Data-driven approaches for quantification of complex biomarkers from medical imaging"

MSc in Biomedical Engineering – Biomechanics and Biomaterials (*cum Laude*)

Apr. 2018

Politecnico di Milano

Milan, Italy

BSc in Biomedical Engineering

Jul. 2015

Politecnico di Milano

Milan, Italy

SKILLS

Programming Libraries	Experienced: python, MATLAB, C. Pytorch, VTK, ITK, FEniCS, elastix	Familiar: C++
Containerization	Docker, Singularity	
Miscellaneous	Experience with HPC systems, including GPU-accelerated clusters	
Simulation software	Ansys, Star-CCM+, OpenFOAM, SimVascular, CRIMSON	
Data processing and visualization	Paraview, Ensignt, 3DSlicer, Solidworks	
Languages	Full proficiency: Italian, English (TOEFL iBT: 107)	

GRANTS & AWARDS

Switch2Product startup competition Dec. 2022
€30,000 [prize](#) for IPSE-XR: A deep learning and mixed reality-based solution to support neurosurgery
Role: core team member

McKusick Fellowship Research Grant May 2023
\$100,000 grant from the [Marfan Foundation](#)
Project title: Machine learning for identification of geometrical and biomechanical markers of aortic dissection in Marfan patients
Role: principal investigator and fellow

PATENTS

- *Metodo e sistema per il perfezionamento di una procedura chirurgica virtuale* (Filed - PCT) - 2023
Method and system for the improvement of a virtual surgical procedure
- *Sistema e metodo per la valutazione morfo-funzionale del distretto vascolare toracico* (Filed - National) - 2023
System and method for the morpho-functional evaluation of the thoracic vascular district.

MOBILITY

Mar. 2022 – Aug. 2022	Visiting PhD student at the University of Cambridge Cambridge Image Analysis Group Host: Prof. Carola-Bibiane Schönlieb
Jan. 2019 – Aug. 2019	Research Engineer at National University of Singapore Cardiovascular Biomechanics and Ultrasound Laboratory Host: Prof. Choon Hwai Yap
May 2018 – Dec 2018	Research Assistant at Imperial College London Biofluids and Transport Group Host: Prof. Xiao Yun Xu
Sep. 2017 – Mar. 2017	Visiting MSc student at Imperial College London Biofluids and Transport Group Host: Prof. Xiao Yun Xu & Prof. Selene Pirola
Aug. 2010 – Jul. 2011	AFS exchange student – Whitefish, MT, USA

TEACHING

2020 – ongoing	Teaching Assistant – Course of Biomechanics – 3rd Year Undergraduate • Preparation of hands-on activities and assignment of group projects • Delivery of tutorials for a total of 4h per week for semester • Exam preparation and evaluation
Jan. 2019 – Aug. 2019	Tutor – Course of Computer Science I (C language) • Supporting students during hands-on activities

SUPERVISION

Currently co-promotor of 7 PhD students. Successfully supervised 2 PhD students, 6 MSc students and 6 BSc students. Among these, 3 MSc students successfully enrolled in PhD studies (at Politecnico di Milano and TU Delft) and published their own work.

PhD Project Co-supervisor	• “A deep learning framework for coronary artery disease risk assessment” – AY 2021 – 2025 (Politecnico di Milano) • “Data-driven methods and applications for 3D transesophageal image analysis” – AY 2021 (Politecnico di Milano)
MSc Thesis Co-supervisor	• “On the behavior of physics informed neural networks in hemodynamic applications” – AY 2022 – 2023 • “Statistical shape modeling of thoracic aortic aneurysm: data-driven synthesis of inlet velocity profiles to overcome clinical data scarcity” – AY 2021 – 2022 • “Mixed reality and deep learning approaches in neurosurgery for craniotomy procedure” – AY 2021 – 2022 • “Blood flow in cerebral aneurysms: comparison among 4D flow guided-CDF, FSI and 3D variational data assimilation” – AY 2019 – 2020 • “Hemodynamic changes due to valve sparing surgery in ascending thoracic aortic aneurysm correction: a tailored multi-patient analysis exploiting the coupled momentum method for fluid-solid interaction” – AY 2019 – 2020 • “Estimation of paravalvular regurgitation associated with TAVI procedures using 4D Flow MRI and aortic root phantoms” – AY 2019 – 2020
BSc Thesis Co-supervisor	• “Segmentation of CT images of abdominal aortic aneurysms to train a deep neural network for automatic segmentation” – AY 2019 – 2020 • “Manual and semi-automatic segmentation from CT images for training a deep neural network for automatic segmentation” – AY 2019 – 2020

PUBLICATIONS

28 peer-reviewed publications, 214 citations, H-index=10 (source: Scopus)

PUBLICATIONS (newest to oldest)

1. R Volleberg, ..., S Saitta, et al., Comprehensive full-vessel segmentation and volumetric plaque quantification for intracoronary optical coherence tomography using deep learning, European Heart Journal – Digital Health (2025) <https://doi.org/10.1093/ehjdh/ztaf021>
2. G Creazzo, G Nannini, **S Saitta**, et al., Deployment of a digital twin using the coupled momentum method for fluid–structure interaction: A case study for aortic aneurysm, Computers in Biology and Medicine (2025) <https://doi.org/10.1016/j.compbiomed.2025.110084>
3. RLM van Herten, ..., S Saitta, I Išgum, World of Forms: Deformable geometric templates for one-shot surface meshing in coronary CT angiography, Medical Image Analysis (2025) <https://doi.org/10.1016/j.media.2025.103582>
4. P Cancian, **S Saitta**, X Gu, et al., Attenuation artifact detection and severity classification in intracoronary OCT using mixed image representations, SPIE Medical Imaging 2025: Image Processing (2025) <https://doi.org/10.1117/12.3047192>
5. I Ianniruberto, F Lo Presi, O Bifulco, D Tondi, **S Saitta**, et al., Ascending aorta over-angulation is a risk factor for acute type A aortic dissection: evidence from advanced finite element simulations, European Journal of Cardio-Thoracic Surgery (2025) <https://doi.org/10.1093/ejcts/ezaf053>
6. M Pillitteri, G Nannini, **S Saitta**, et al., Pericoronary adipose tissue attenuation as a predictor of functional severity of coronary stenosis, arXiv preprint (2025) <https://doi.org/10.48550/arXiv.2502.13649>
7. G Nannini, J Suk, P Rygiel, **S Saitta**, et al., Learning Hemodynamic Scalar Fields on Coronary Artery Meshes: A Benchmark of Geometric Deep Learning Models, Computers in Biology and Medicine (2025) <https://doi.org/10.1016/j.compbiomed.2025.110477>

8. R Munafò, **S Saitta**, D Tondi, et al., Automatic 4D mitral valve segmentation from transesophageal echocardiography: a semi-supervised learning approach, *Medical & Biological Engineering and Computing* (2025) <https://doi.org/10.1007/s11517-024-03275-w>
9. R Munafò, **S Saitta**, L Vicentini, et al., MitraClip Device Automated Localization in 3D Transesophageal Echocardiography via Deep Learning, *arXiv preprint* (2024) <https://doi.org/10.48550/arXiv.2412.15013>
10. G Nannini, **S Saitta**, L Mariani, et al., An automated and time-efficient framework for simulation of coronary blood flow under steady and pulsatile conditions, *Computer Methods and Programs in Biomedicine* (2024) <https://doi.org/10.1016/j.cmpb.2024.108415>
11. A Riva, **S Saitta**, F Sturla, et al., Left ventricle diastolic vortex ring characterization in ischemic cardiomyopathy: insight into atrio-ventricular interplay, *Medical & Biological Engineering & Computing* (2024) <https://doi.org/10.1007/s11517-024-03154-4>
12. R Gorla, O Oliva, L Arzuffi, V Milani, **S Saitta**, et al., Angulation and curvature of aortic landing zone affect implantation depth in transcatheter aortic valve implantation, *Scientific Reports* (2024) <https://doi.org/10.1038/s41598-024-61084-5>
13. G Nannini, **S Saitta**, A Baggiano, et al., A fully automated deep learning approach for coronary artery segmentation and comprehensive characterization, *APL Bioengineering* (2024) <https://doi.org/10.1063/5.0181281>
14. R Munafò, **S Saitta**, G Ingallina, P Denti, F Maisano, et al., A Deep Learning-Based and Fully Automated Pipeline for Regurgitant Mitral Valve Anatomy Analysis from 3D Echocardiography, *IEEE Access* (2024) <https://doi.org/10.48550/arXiv.2302.10634>
15. **S Saitta**, M Carioni, S Mukherjee, CB Schönlieb, A Redaelli. Implicit neural representations for unsupervised super-resolution and denoising of 4D flow MRI. *Computer Methods and Programs in Biomedicine* (2024) <https://doi.org/10.48550/arXiv.2302.12835>
16. **S Saitta**, F Sturla, R Gorla, OA Oliva, E Votta, F Bedogni, A Redaelli, A CT-based deep learning system for automatic assessment of aortic root morphology for TAVI planning, *Computers in Biology and Medicine* (2023) <https://doi.org/10.1016/j.compbiomed.2023.107147>
17. **S Saitta**, L Maga, C Armour, E Votta, DP O'Regan, MY Salmasi, et al., Data-driven generation of 4D velocity profiles in the aneurysmal ascending aorta, *Computer Methods and Programs in Biomedicine* (2023) <https://doi.org/10.1016/j.cmpb.2023.107468>
18. P Aigner, E Sella Bart, S Panfili, T Körner, M Mach, M Andreas, M Königshofer, **S Saitta**, et al. Quantification of paravalvular leaks associated with TAVI implants using 4D MRI in an aortic root phantom made possible by the use of 3D printing, *Frontiers in Cardiovascular Medicine* (2023) <https://doi.org/10.3389/fcvm.2023.1083300>
19. A Riva, F Sturla, S Pica, A Camporeale, L Tondi, **S Saitta**, et al. Comparison of four-dimensional magnetic resonance imaging analysis of left ventricular fluid dynamics and energetics in ischemic and restrictive cardiomyopathies, *Journal of Magnetic Resonance Imaging* (2022) <https://doi.org/10.1002/jmri.28076>
20. MC Palumbo*, **S Saitta***, M Schiariti, MC Sbarra, et al., Mixed Reality and Deep Learning for External Ventricular Drainage Placement: A Fast and Automatic Workflow for Emergency Treatments, *MICCAI* (2022) *equal contribution https://doi.org/10.1007/978-3-031-16449-1_15
21. C Armour, B Guo, **S Saitta**, et al., The role of multiple re-entry tears in Type B aortic dissection progression: a longitudinal study using a controlled swine model, *Journal of Endovascular Therapy* (2022) <https://doi.org/10.1177/15266028221111295>
22. **S Saitta**, F Sturla, A Caimi, et al., A deep learning-based and fully automated pipeline for thoracic aorta geometric analysis and planning for endovascular repair from computed tomography, *Journal of Digital Imaging* (2022) <https://doi.org/10.1007/s10278-021-00535-1>
23. CH Armour, B Guo, **S Saitta**, et al., Evaluation and verification of patient-specific modelling of type B aortic dissection, *Computers in Biology and Medicine* (2022) <https://doi.org/10.1016/j.compbiomed.2021.105053>
24. **S Saitta**, B Guo, S Pirola, et al., Qualitative and quantitative assessments of blood flow on tears in type B aortic dissection with different morphologies, *Frontiers in Bioengineering and Biotechnology* (2021) <https://doi.org/10.3389/fbioe.2021.742985>
25. G Nannini, A Caimi, MC Palumbo, **S Saitta**, et al., Aortic hemodynamics assessment prior and after valve sparing reconstruction: A patient-specific 4D flow-based FSI model, *Computers in Biology and Medicine* (2021) <https://doi.org/10.1016/j.compbiomed.2021.104581>

26. CH Armour, B Guo, S Pirola, **S Saitta**, et al., The influence of inlet velocity profile on predicted flow in type B aortic dissection, *Biomechanics and modeling in mechanobiology* (2021) <https://doi.org/10.1007/s10237-020-01395-4>
27. **S Saitta**, S Pirola, F Piatti, E Votta, et al., Evaluation of 4D flow MRI-based non-invasive pressure assessment in aortic coarctations, *Journal of Biomechanics* (2019) <https://doi.org/10.1016/j.jbiomech.2019.07.004>
28. S Pirola, B Guo, C Menichini, **S Saitta**, et al., 4-D flow MRI-based computational analysis of blood flow in patient-specific aortic dissection, *IEEE Transactions on Biomedical Engineering* (2019) [10.1109/TBME.2019.2904885](https://doi.org/10.1109/TBME.2019.2904885)

PROCEEDINGS (Selected – first author podium presentations only)

1. **S Saitta**, M Carioni, et al., Implicit neural representations for unsupervised denoising and super-resolution of 4D flow MRI, *ICIAM* (2023) – Invited podium presentation
2. **S Saitta**, F Sturla, et al., A CT-based deep learning system for automatic assessment of aortic root morphology for TAVI planning, *ESB* (2023) – Podium presentation
3. **S Saitta**, R Munafò, et al., Hemodynamic assessment in abdominal aortic aneurysm: comparison among 4D flow MRI, CFD and 4DVar, *GIMC SIMAI* (2022) – Invited podium presentation
4. **S Saitta**, R Munafò, E Votta, Fast automatic segmentation of mitral valve structures from 3D transesophageal echocardiography for transcatheter procedures: training and validation of a 3D U-Net convolutional neural network, *ECCOMAS* (2022) – Invited podium presentation
5. **S Saitta**, F Sturla, A Caimi, et al., A deep learning-based and fully automated pipeline for thoracic aorta geometric analysis and TEVAR planning from computed tomography, *EACVI* (2021) – Podium presentation